

DAWSON COMMUNITY AP, MT, USA

WMO: 726676

Lat: 47.133N

Lon: 104.800W

Elev: 749

StdP: 92.65

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24087

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-23.9	-31.9	0.2	-26.3	-28.1	0.3	-23.4	14.7	-2.8	13.2	-3.0	3.3	320	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	34.8	18.7	32.8	18.3	31.2	18.0	21.1	30.0	20.0	29.0	19.1	28.1	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	14.1	24.7	17.2	13.5	24.0	16.0	12.5	23.1	64.6	30.0	60.5	28.7	56.9	28.2	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	11.4	9.9	-31.3	38.1	3.5	1.8	-33.8	39.4	-35.8	40.4	-37.8	41.4	-40.3	42.7
			-30.8	23.0	2.4	1.3	-32.5	24.0	-33.9	24.8	-35.2	25.5	-36.9	26.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-8.5	-6.7	0.1	6.2	11.8	17.4	22.3	21.1	15.3	7.2	-0.1	-6.6		
	DBStd	12.16	8.17	8.13	7.65	5.39	4.65	3.98	3.34	3.76	5.08	5.27	6.60	7.51		
	HDD10.0	2467	572	467	311	133	34	1	0	0	12	117	306	514		
	HDD18.3	4540	831	699	564	363	206	63	6	17	117	347	554	773		
	CDD10.0	1263	0	0	4	20	91	223	380	343	171	29	2	0		
	CDD18.3	296	0	0	0	0	4	36	127	102	26	1	0	0		
	CDH23.3	4079	0	0	2	20	104	499	1636	1353	433	33	0	0		
	CDH26.7	1784	0	0	0	4	22	172	769	633	177	8	0	0		
(14) Wind	WSAvg	4.5	4.5	4.4	4.8	5.0	4.9	4.6	4.2	4.2	4.3	4.6	4.4	4.4		
(15) Precipitation	PrecAvg	360	10	9	13	34	60	68	57	31	30	30	12	10		
	PrecMax	498	30	30	56	94	168	135	173	70	125	89	59	26		
	PrecMin	244	1	0	1	10	14	25	9	3	4	9	0	0		
	PrecStd	63	7	8	12	22	39	32	41	17	27	20	14	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	13.3	22.1	26.3	29.1	34.1	37.4	37.3	34.2	27.4	18.7	11.1		
		MCWB	4.9	6.4	10.5	12.8	16.6	19.5	19.4	19.1	17.6	14.2	9.4	4.9		
	2%	DB	6.4	9.0	17.4	22.2	26.3	31.0	35.0	34.3	31.3	22.4	14.1	7.4		
		MCWB	2.7	4.2	8.4	11.3	15.3	18.9	19.1	18.2	16.7	12.1	6.8	2.9		
	5%	DB	3.9	6.3	13.8	18.7	23.7	28.1	32.9	32.5	28.0	18.8	11.3	4.9		
		MCWB	1.2	2.3	6.5	9.8	13.8	17.7	18.9	17.9	15.4	10.4	5.6	1.2		
	10%	DB	2.5	3.7	11.0	16.2	21.3	26.2	31.3	30.9	25.2	16.1	8.6	2.7		
		MCWB	0.2	0.9	5.3	8.1	12.7	16.5	18.6	17.7	14.4	9.0	4.1	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	7.0	10.8	13.8	18.4	21.8	23.2	21.6	19.6	15.1	10.1	5.2		
		MCDB	9.0	12.7	20.2	24.5	25.2	30.0	32.0	31.1	29.3	24.9	17.9	9.7		
	2%	WB	2.9	4.5	8.9	11.7	16.2	20.2	21.6	20.2	17.9	12.7	7.7	3.0		
		MCDB	6.2	8.8	16.7	21.2	24.1	28.1	30.5	29.9	27.3	20.2	14.0	7.0		
	5%	WB	1.5	2.7	7.1	10.1	14.8	18.8	20.4	19.2	16.6	11.0	5.8	1.6		
		MCDB	4.1	6.0	13.5	17.7	22.2	26.6	29.3	28.9	26.3	18.1	10.8	4.7		
	10%	WB	0.2	1.0	5.3	8.7	13.3	17.4	19.5	18.3	15.2	9.5	4.1	0.2		
		MCDB	2.4	3.8	10.1	15.4	20.0	24.8	28.5	28.0	24.4	15.4	8.0	2.7		
(35) Mean Daily Temperature Range	MDBR		10.4	11.0	12.3	13.1	12.8	13.2	14.8	15.2	14.7	12.6	11.6	10.2		
	5% DB	MCDBR	11.3	13.5	17.5	18.4	16.9	16.9	18.0	18.5	19.3	17.9	15.8	12.3		
		MCWBR	8.5	9.3	10.3	9.9	8.5	7.8	6.8	7.1	8.6	9.5	9.8	8.7		
	5% WB	MCDBR	11.1	13.1	16.8	16.8	15.0	15.5	15.6	16.0	17.1	16.5	15.3	12.0		
		MCWBR	8.5	9.2	10.2	9.4	8.0	7.8	6.8	6.9	7.9	9.1	9.7	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.257	0.274	0.309	0.331	0.348	0.355	0.359	0.319	0.288	0.260	0.245		
	taud		2.532	2.553	2.561	2.456	2.454	2.444	2.445	2.403	2.497	2.572	2.550	2.537		
	Ebn at Noon		873	939	965	949	933	915	902	885	897	884	850	838		
	Edh at Noon		60	72	84	103	108	110	108	108	89	71	58	53		
(44) All-Sky Solar Radiation	RadAvg		1.58	2.56	3.77	4.94	5.87	6.67	7.12	5.85	4.32	2.76	1.72	1.23		
	RadStd		0.13	0.19	0.28	0.28	0.55	0.42	0.26	0.35	0.33	0.23	0.18	0.06		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page